

GOPALAKRISHNAN PALANISAMY

Software Development Engineer in Test (SDET)

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PROFESSIONAL SUMMARY

Results-driven SDET with 8+ years of overall IT experience, including 6+ years as an SDET designing and delivering scalable test automation frameworks for Web, Mobile (Android / iOS / mSite), and API layers. Possesses 3+ years of hands-on experience in **Flutter Mobile and Web automation testing** (Android & iOS) using Appium, with strong expertise in **YAML-based configuration management for Flutter automation frameworks**. Expert in Selenium WebDriver, Appium, Java, Python (pytest), Rest Assured, and Playwright with deep knowledge of POM, Hybrid, and Data-Driven frameworks. Proven record of up to 99% reduction in manual testing effort. Skilled in BDD (Cucumber), CI/CD (Jenkins / GitHub Actions), cross-browser and cross-platform mobile testing, and Agile/Scrum.

CORE TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript

Test Automation: Selenium WebDriver, Appium, Playwright

Mobile Testing: Android (Native & Flutter), iOS (Native & Flutter), mSite / Responsive Web

Python Testing: pytest, pytest-html, pytest-xdist, pytest-bdd, requests

API Testing: Postman, Rest Assured, Python requests

Frameworks & Patterns: TestNG, Cucumber (BDD), pytest, POM, Hybrid Framework, Data-Driven Testing

CI/CD & Version Control: Jenkins, GitHub Actions, Git, GitHub

YAML & Config Management: Flutter automation framework configs, device capabilities, execution profiles, CI/CD pipeline YAML

Test & Defect Mgmt: JIRA, Test Planning, Test Case Design, Defect Lifecycle Management

Database & Validation: SQL, MySQL, Azure Cosmos DB, Apache Kafka (Message Validation)

AI Tools: Claude (custom framework for gap analysis), ChatGPT (OpenAI), GitHub Copilot, Google Gemini

PROFESSIONAL EXPERIENCE

Software Development Engineer in Test (SDET) | Moolya Software Testing Pvt Ltd, Bengaluru, India | Oct 2024 – Present

- Developed and maintained end-to-end UI automation scripts using Playwright with JavaScript/TypeScript for cross-browser web application testing.
- Implemented Playwright-based automation frameworks with reusable components, locator strategies, and asynchronous handling using JavaScript/TypeScript.
- Designed data-driven and BDD-based automation scenarios using Playwright with TypeScript, improving test maintainability and execution efficiency.
- Utilized Playwright features such as auto-waits, network interception, screenshots, tracing, and parallel execution for robust automation validation.
- Automated 170+ test cases across Web and Mobile using TestNG and Cucumber (BDD), reducing manual testing effort by 65% and accelerating release cycles.
- Engineered an OCR-based automation solution for banner and PLP discount validation — validating product counts, discount ranges, and business rules with automated Excel reporting, cutting validation time from 1 week (3 engineers) to under 1 hour (~99% effort reduction).
- Integrated automation suites into Jenkins CI/CD pipelines with GitHub Actions for continuous test execution triggered on every build deployment.
- Developed Python-based test automation scripts using pytest framework for backend API validation and regression testing, integrating pytest-html for automated HTML report generation.
- Designed reusable pytest fixtures and parameterized test cases using `@pytest.mark.parametrize` for data-driven testing across multiple test environments, reducing test script maintenance overhead by 35%.
- Performed code reviews, enforced automation best practices, and mentored junior SDETs, improving framework stability and maintainability.
- Leveraged GitHub Copilot and ChatGPT to accelerate test script development and reduce boilerplate effort.

Software Test Engineer | Client: Mobius Knowledge Services, Chennai, India | Oct 2022 – Oct 2024

- Led API automation strategy using Rest Assured and Postman, covering 100+ API scenarios across functional, negative, and regression testing.
- Reduced production defects by 50% through early API contract validation and proactive defect triage.
- Built Python pytest automation scripts for API regression and functional testing, leveraging the requests library to complement existing Rest Assured test suites, achieving comprehensive end-to-end test coverage across microservices.
- Utilized pytest-xdist plugin for parallel test execution and pytest-html for structured reporting, reducing overall test execution time by 25% and improving stakeholder visibility into quality metrics.
- Improved test execution efficiency by 30% via optimized test data management and parallel execution.
- Actively contributed to test planning, backlog grooming, and sprint quality readiness.
- Developed reusable custom Playwright helper utilities for browser initialization, API mocking, screenshot capturing, tracing, reporting, retry handling, and parallel execution to improve framework stability and maintainability.
- Conducted root cause analysis for flaky test failures by analyzing Playwright traces, screenshots, execution logs, console logs, and network activity to improve automation reliability.

Business Intelligence Engineer | OEC Services India Pvt Ltd, Chennai, India | Feb 2022 – Sep 2022

- Validated event-driven data pipelines using Apache Kafka, verifying message payloads, and ensuring downstream data persistence in Azure Cosmos DB through ETL validation.
- Conducted SQL-based ETL validations, data integrity checks, and end-to-end data flow testing.
- Conducted backend API validation for data ingestion services ensuring correct payload structure.

Software Test Engineer | Client: Prodapt Solutions, Chennai, India | Dec 2019 – Feb 2022

- Developed and maintained 150+ automated API test cases using Rest Assured with Java, increasing defect detection rate by 30% and improving regression coverage.
- Managed full defect lifecycle in JIRA — logging, tracking, verification, closure — with daily execution reporting to stakeholders.
- Built and maintained a data-driven automation framework enabling parameterized test execution across multiple datasets, improving framework reusability.
- Actively participated in SDLC and STLC — test planning, test design, and execution sign-off across all release milestones.

KEY ACHIEVEMENTS AND IMPACT

- **99% effort reduction:** OCR-based automation slashed banner/PLP discount validation from 1 week (3 engineers) to under 1 hour.
- **65% manual effort reduction:** Automated 170+ Web and Mobile test cases at Moolya (including Flutter Android/iOS), significantly accelerating release velocity.
- **50% fewer production defects:** Early-stage API defect detection at Mobius reduced post-release issues by half.
- **35% reduction in test maintenance:** Python pytest fixtures and parameterized test design at Moolya improved reusability and cut upkeep overhead.
- **30% higher defect detection rate:** 150+ Rest Assured API test cases at Prodapt strengthened regression coverage significantly.
- **AI-driven gap coverage:** Custom Claude AI framework integration enabled systematic identification of test coverage gaps, reducing missed scenarios across Flutter mobile and web platforms.

TESTING METHODOLOGIES AND PRACTICES

Agile / Scrum | SDLC and STLC | Shift-Left Testing | BDD (Cucumber) | Data-Driven Testing | POM | Hybrid Framework Design | Flutter Mobile & Web Automation | iOS & Android Cross-Platform Testing | YAML Config Management | Cross-Browser Testing | Regression Testing | Smoke & Sanity Testing | API Contract Testing | Functional & Integration Testing | Negative Testing | Python pytest | GitHub Copilot | ChatGPT | Google Gemini

EDUCATION

Bachelor of Science – Information Technology

AG Arts and Science College, Tirupur, Tamil Nadu, India | 2017